

ALIA SAJANI

alia.a.sajani.gr@dartmouth.edu | 832-472-9747

EDUCATION

Dartmouth College, Guarini School of Graduate and Advanced Studies, Hanover, NH

Ph.D. Candidate, Immunology, August 2018 – present
Anticipated January 2024

Grinnell College, Grinnell, IA

Bachelor of Arts, Biology, August 2012 – May 2016

Grinnell-in-London and IES Abroad, London, United Kingdom

Study Abroad Experience, August 2014 – December 2014

RESEARCH EXPERIENCE

Dartmouth School of Medicine, Hanover, NH

Co-PIs: Drs. Randolph Noelle and Mary Jo Turk

Graduate Student Researcher, August 2018-present

- Describing naive CD4⁺ T cell heterogeneity and differentiation using single cell sequencing technologies, as well as in vivo and in vitro mouse models
- Complementing immunology research techniques such as fluorescence-activated cell sorting (FACS), adoptive transfer experiments and in vitro cell culturing with computational analyses of bulk and single cell transcriptional datasets
- Exploring how the immune checkpoint VISTA alters CD8⁺ T cell peripheral tolerance

National Institutes of Health, National Institute of Allergy and Infectious Diseases, Immunopathogenesis Section, Bethesda, MD

Postbaccalaureate Intramural Research Training Awardee, August 2016-July 2018

PI: Dr. Anthony Fauci

- Investigated the cellular and molecular mechanisms of immune dysfunction during HIV infection
- Evaluated the role of MAdCAM-1 and integrin $\alpha_4\beta_7$ in the activation of CD4⁺ T cells, which enhances HIV replication
- Analyzed receptor cell surface expression after activation, as well as cell phenotype, using flow cytometry

Mayo Clinic, Department of Immunology, Rochester, MN

PI: Dr. Karen Hedin

Summer Undergraduate Research Fellow, May-August 2015

- Characterized TCR mediated down-regulation of CXCR4 surface expression by studying receptor cell-surface regulation
- Analyzed a signal transduction pathway using kinase-deficient cells and drug inhibition of specific kinases
- Enrolled in an in-depth immunology class hosted at Mayo Clinic during the summer

Grinnell College, Department of Biology, Grinnell, IA

PI: Dr. Laura Burrack

Research Assistant, August 2013-May 2014

- Assessed the role of neocentromeres in kinetochore formation in *C. albicans*
- Completed restocking, equipment preparation, and autoclaving to maintain a clean and functional lab environment

PEER-REVIEWED PUBLICATIONS

Dina Rabadi, Alia Sajani, Randolph J. Noelle, and J. Louise Lines (2022). The role of VISTA in the tumor microenvironment. *Journal of Cancer Metastasis and Treatment*. <https://dx.doi.org/10.20517/2394-4722.2022.06>

Livia R. Goes, Alia Sajani, Aida Sivo, Ronke Olowojesiku, Jocelyn C. Ray, Ian Perrone, Jason Yoltz, Alexandre Girard, Louise Leyre, Constantinos Kurt Wibmer, Lynn Morris, Giacomo Gorini, Genoveffa Franchini, Rosemarie D. Mason, Mario Roederer, Saurabh Mehandru, Marcelo A. Soares, Claudia Cicala, Anthony S. Fauci, and James Arthos (2020). The V2 loop of HIV gp120 delivers costimulatory signals to CD4⁺ T cells through Integrin $\alpha_4\beta_7$ and promotes cellular activation and infection. *Proceedings of the National Academy of Sciences*, <https://doi.org/10.1073/pnas.2011501117>

Fatima Nawaz, Livia R. Goes, Jocelyn C. Ray, Ronke Olowojesiku, Alia Sajani, Aftab A. Ansari, Ian Perrone, Joseph Hiatt, Donald Van Ryk, Danlan Wei, Mia Waliszewski, Marcelo A. Soares, Katija Jelacic, Mark Connors, Stephen A. Migueles, Elena Martinelli, Francois Villinger, Claudia Cicala, Anthony S. Fauci, and James Arthos (2018). MAdCAM costimulation through Integrin- $\alpha_4\beta_7$ promotes HIV replication, *Mucosal Immunology*, <https://doi.org/10.1038/s41385-018-0044-1>

PUBLICATIONS IN PROGRESS

Alia Sajani, Mohamed ElTanbouly, Evelien Schaafsma, Chao Chang, J. Louise Lines, Mary Jo Turk, and Randolph J. Noelle. The immune environment dynamically controls naive CD4⁺ T cell heterogeneity.

RECENT PRESENTATIONS

Alia Sajani, Mohamed ElTanbouly, Evelien Schaafsma, Chao Chang, Kris Hogquist, Stephen Jameson, J. Louise Lines, Mary Jo Turk, and Randolph J. Noelle. Naive CD4⁺ T Cell heterogeneity. Presented at the Dartmouth Immunology Department Retreat in Woodstock, VT, 25-26 April 2022 and 8-9 May 2023.

Alia Sajani, Mohamed ElTanbouly, Evelien Schaafsma, Chao Chang, Kris Hogquist, Stephen Jameson, J. Louise Lines, Mary Jo Turk, and Randolph J. Noelle. Naïve CD4⁺ T Cell heterogeneity. Presented at the American Association of Immunologists Conference in Portland, Oregon, 6-10 May 2022.

SCIENCE WRITING AND EDITING

National Association for Science Writers

Contributing Writer, June 2021-present

- Wrote articles on the intersection of science and society, formatted to journalistic standards such as AP style, accompanied with fact checking report
- Conducted interviews of scientists and collaborated with editors for publications
- Participated in the David Perlman Summer Mentoring Program as well as the ScienceWriters2021 conference
- Press for the 2022 American Association for the Advancement of Science annual meeting

Immunobites

Writer and Editor, October 2020-present

- Contributed to the development and publication of articles on recent topics in immunology research for a general audience
- Trained on best practices for editing
- Created illustrations using BioRender to better explain complex scientific concepts

Science Journalism Forum

Participant, September 2021

- Received the SJF2021 fellowship to attend the conference and learn about science communication

StoryCollider

Participant, August 2021

- Engaged in a week-long workshop on best practices for science storytelling
- Critiqued example pieces to determine what makes a good story
- Wrote short stories about science and received feedback from instructors and other workshop participants

National Institutes of Health, The NIH Catalyst, Bethesda, MD

Writer, March 2017- July 2018

- Wrote research briefs on NIH funded research discoveries
- Coached by editor on scientific journalism practices from recording interviews to editing, leading to publications

MRC National Institute for Medical Research (Francis Crick Institute), London, United Kingdom

Science Writing Intern, August-December 2014

- Wrote news items for the NIMR website to summarize recently published research articles by NIMR researchers
- Published biographical entries for past NIMR scientists on the history blog
- Updated and created Wikipedia pages on prominent minority scientists from the NIMR's history
- Cumulative project on the NIMR World Health Organization's response to the 2009 H1N1 epidemic

PUBLISHED SCIENCE WRITING

Finding Love in a Hopeless Place: How deep-sea anglerfish evolved to fuse with their mates, *Immunobites*

Could social media help reverse the mistrust of science?, *NASW*

Anthropologists push for policy changes to undo the erasure of the "invisible" dead, *NASW*

Making the Invisible Visible: Challenges to explaining deep tech, *NASW*

T cells as the antidote for a broken heart, *Immunobites*

History of the National Library of Medicine, *NIH Catalyst*

HONORS AND AWARDS

National Institute of Allergy and Infectious Diseases INRO Scholar

NIH Postbaccalaureate Poster Day Outstanding Poster Award

Science Journalism Forum Fellowship

Florence Sifferd Science Scholarship (Grinnell College)

MCB Fellowship (Dartmouth College)

Global Health Fellowship (Dartmouth College)

LEADERSHIP, ACTIVITIES AND COMMUNITY SERVICE

Representative, Graduate Women in Science & Engineering, Dartmouth College, Hanover, NH

August 2021- May 2022

Global Health Fellow, Dartmouth College, Hanover, NH

August 2020-May 2021

Benchling Campus Ambassador, Dartmouth College, Hanover, NH

October 2019-May 2020

Genetics Graduate Teaching Assistant, Dartmouth College, Hanover, NH

January-March 2020

Mentor, Cientifico Latino, Graduate Program Application Mentor

September 2019-May 2020

Executive Committee , Dartmouth Writers Society, Hanover, NH	June 2019-May 2020
Poster Judging Committee Member , Diocesan Middle School Science Fair, Fairfax, VA	April 2017
Student , Fiction Writing Workshop, George Washington University, Washington, DC	January-May 2017
Review Panel Member , Grinnell Young Innovator for Social Justice Prize, Grinnell, IA	September 2013-May 2016
Dancer , Grinnell College Dance Ensemble, Grinnell, IA	August-December 2015
Student Science Teacher , Grinnell-Newburg Davis Community School, Grinnell, IA	April 2015
Co-Leader , Spring Break Youth Enrichment Through Dance Service Trip, Atlanta, GA	March 2015
Volunteer , Spring Break Service Trip, Habitat for Humanity Flood Relief, Cedar Rapids, IA	March 2014
Volunteer , Spring Break Service Trip, Refugee and Immigrant Institute, St Louis, MO	March 2013
Member , Social Entrepreneurs of Grinnell, Grinnell, IA	August 2012-May 2016
Member , Grinnell Science Project, Grinnell, IA	August 2012-May 2016

PROFESSIONAL ASSOCIATIONS

National Association of Science Writers 2021-present
Federation of Clinical Immunology Societies 2019-present

LANGUAGES

Swahili (native proficiency)
Gujarati (native proficiency)
Kutchi (native proficiency)
Spanish (proficiency)
French (limited proficiency)
Hindi (limited proficiency)
Urdu (limited proficiency)